Module 2

Fundamentals of Nutrition

Malaysian Food Pyramid 2020 and Malaysian Healthy Plate
Understanding of Macronutrients, Micronutrients, Phytonutrients and Hydration

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Learning Objectives

By the end of this module, you will be able to:

- Understand the concept of balance, moderation, variety and adequate in eating
- Apply the Malaysian Food Pyramid 2020 and Malaysian Healthy Plate (Quarter-Quarter-Half) in daily eating
- Understand the importance of macronutrients and micronutrients
- Understand phytonutrients and their role in health

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Does your meal look like this?





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Key Concepts: Balance, Moderation, Variety & Adequate

Balance	Moderation	Variety	Adequate
A combination of foods that provides a balance of nutrients	Eat the right amount of food as per the recommended number of servings	Eat a variety of foods within each level of food pyramid	Enough energy, nutrients and fibre to maintain general health
Eat the right amounts from all food groups	Avoid overeating, especially foods high in sugar, salt and fat	Variety ensures you get all essential nutrients and phytonutrients	Individual plan
No single group provides all nutrients — different food groups complement each other			

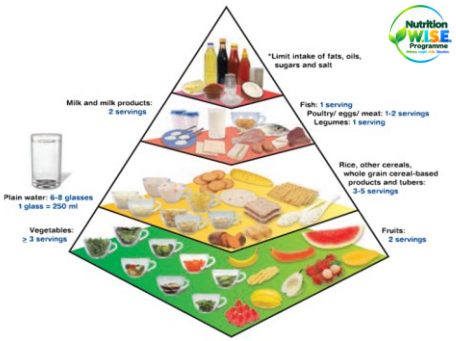
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Malaysian Food Pyramid 2020

Choose your daily food intake based on the recommended number of servings in the Malaysian Food Pyramid 2020:

Serve age-appropriate portions



- The number of servings is calculated based on 1500 to 2300 kcal.
- This pyramid is meant for children aged 7 years and older; for younger children, refer to the Malaysian Dietary Guidelines (MDG) for Children and Adolescents.
- For adolescents aged 13 to 15 years, the recommendation for fruits is 2-3 servings and for milk and milk products 2-3 servings.
- For adolescents aged 16 to < 18 years, the recommendation for fruits is 2-3 servings, milk and milk products 2-3 servings and for rice, other cereals, whole grain cereal-based products and tubers 3-6 servings.
- This includes ultra-processed foods which contain artificial substances such as colours, sweeteners, flavours, preservatives, and other additives.

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Malaysian Healthy Plate: Quarter-Quarter-Half

Visual guide for balanced meal portions at each meal

10 inches (25 cm)

Equivalent to one palm size

¼ Plate: Protein sources
Fish, chicken, eggs, legumes, tofu

¼ Plate: Cereals
Wholegrains, rice, bread, noodles

Equivalent to one fist size



½ Plate: Vegetables and fruits
Emphasise vegetables; include a variety of colour

Equivalent to two cupped hands

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How much to eat?

Macronutrients

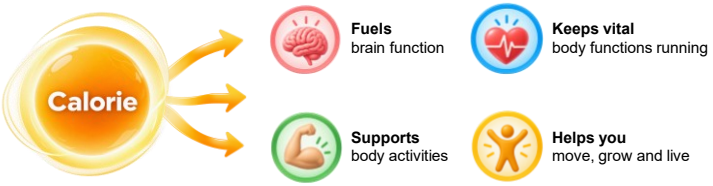


Micronutrients

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What is calorie/energy?

A calorie is a unit used to measure the amount of **energy** provided by your food and drinks



In short:
Calories give your body the energy it needs to function.

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What are macronutrients?

Daily energy requirements:

Adult (18-59 years old), sedentary lifestyle

Men:
2,000 kcal/day

Women:
1,500 kcal/day

Varies with:

Age

Activity level

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Spot the difference

Calorie

A measure of **energy** provided by food and drinks.



Orange

- ✓ Energy
- ✓ Fibre
- ✓ Vitamin C
- ✓ Potassium
- ✓ Phytonutrients

Empty Calorie

Calories that provide **little or no nutritional value**



Orange-flavoured soft drink

- ✓ Energy

Calorie ≠ bad
Think: Energy + Nutrients

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What are macronutrients?

- Nutrients required by the body in LARGE amounts
- Provide energy and are essential for growth, metabolism and overall body functions



Carbohydrates

Primary source of energy
4 kcal/gram



Protein

Build and repair tissues
4 kcal/gram



Fats

Stored energy
9 kcal/gram

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Carbohydrates



Main source of energy for the brain & body

Broken down into glucose for immediate use

Recommended daily intake:

3-5 servings
Rice, other cereals, whole grain cereal-based products, & tubers

≥3 servings
Vegetables

2 servings
Fruits

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Carbohydrates

Macronutrients

Simple carbohydrate

Monosaccharide → **Disaccharide**

Mono = 1 sugar unit
Glucose, fructose, galactose

Di = 2 sugar units
Sucrose, lactose, maltose

Fruits, milk, sweets

Complex carbohydrate

Polysaccharide

Poly = many sugar units
Starch, glycogen, fibre

Brown rice, wholemeal bread, legumes, vegetables

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Carbohydrates

Macronutrients

Complex carbohydrate

Polysaccharide

Poly = many sugar units
Starch, glycogen, fibre

Brown rice, wholemeal bread, legumes, vegetables

Why more complex carbohydrates?

- ✓ Sustained energy supply: Steady stream of energy
- ✓ Blood sugar regulation: Maintain balanced blood sugar levels
- ✓ Improve digestive health: Rich in dietary fibre
- ✓ Weight management support: Slow digestion, feel full longer

Refined grain → whole grain
Sugary snack → fruit + nuts
Sweetened drink → plain water

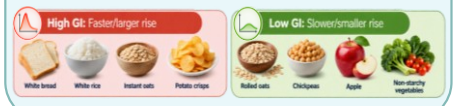
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Carbohydrates

Macronutrients

Blood sugar regulation: Glycaemic Index (GI) & Glycaemic Load (GL)

Glycaemic Index (GI)
= Speed
How quickly a carbohydrates raises blood glucose levels



Glycaemic Load (GL)
= Amount
Takes into account both the GI and the amount of carbohydrate in a serving

$$GL = \frac{GI \times \text{grams of carbohydrate per serving}}{100}$$

A low-GI food can still contribute a large glucose load if the portion is large.

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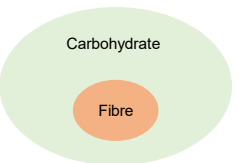
Carbohydrates

Macronutrients

Dietary fibre:

A plant-based carbohydrate that human enzymes cannot digest or absorb

But… not all carbohydrates are fibre



Why do we need fibre?

- Supports digestive health
- Helps control blood glucose & cholesterol levels

Recommended daily intake (adult):
20-30 g of dietary fibre

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Carbohydrates

Macronutrients

Soluble fibre

Dissolve/disperse in water – may form a gel

- Slows digestion
- Lowers cholesterol
- Feeds gut bacteria



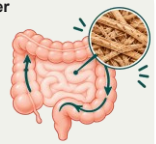
Apple, banana, oats, pea, barley



Insoluble fibre

Does not dissolve in water

- Prevents constipation
- Supports bowel health



Wheat bran, nuts, potato with skin, whole grains



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Protein

Macronutrients

- Builds new cells
- Repairs old cells
- Muscle growth
- Makes enzymes & hormones



Sources

Animal:
fish, shellfish, meat, poultry, egg, milk

Plant:
Beans, peas, soy & soy products

Recommended intake:

- 1 serving Fish
- 1-2 servings Poultry/eggs/meat
- 1 serving Legumes
- 2 servings Milk & milk products

General recommendation for adult:
0.8 – 1.0 g of protein per body weight (kg)

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Protein

Macronutrients

Primary → **Secondary** → **Tertiary** → **Quaternary**



A single chain of amino acids linked in sequence



The chain folds locally into coils/pleated sheets



The whole chain folds into a compact 3D shape



Two or more folded chains assemble into one functional protein

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Protein

Macronutrients

Essential amino acids (9)

Cannot be synthesised by the body

Histidine	Isoleucine	Leucine
Lysine	Methionine	Phenylalanine
Threonine	Tryptophan	Valine

Non-essential amino acids (11)

The body can synthesise

Alanine	Arginine*	Asparagine
Aspartic acid	Cysteine*	Glutamic acid
Glutamine*	Glycine*	Proline*
Serine *	Tyrosine*	

* Conditionally essential in certain life stages or illnesses

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Protein

Macronutrients

Complete Protein

Contains all 9 essential amino acids in adequate amounts

Mostly in animal-based foods & a few plants



Food sources

Beef, Chicken, Fish, Egg, Dairy, Soy

Incomplete Protein

Missing or low in at least one essential amino acid

Mostly in plant-based foods



Food sources

Beans, Lentils, Nuts, Seeds, Whole grains

Protein complementation:
Combine different plant proteins across the diet

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Protein

Macronutrients

Measuring protein quality:

Protein Digestibility-Corrected Amino Acid Score (PDCAAS)

- ✓ Essential amino acid content and how well the body can digest and absorb it
- ✓ From 0.0 (very poor quality) to 1.0 (high quality)

Protein Source	PDCAAS
Casein	1.0
Whey protein	1.0
Egg	1.0
Soy protein isolate	1.0
Beef	0.92
Pea protein isolate	0.89
Rice protein	0.50
Wheat glutein	0,24

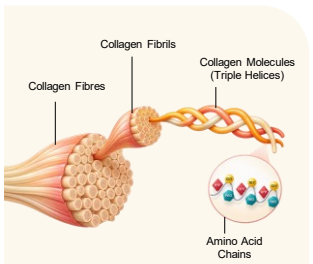
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Protein

Macronutrients

Collagen

The most abundant protein in the human body



30%

Makes up roughly 30% of all protein in the human body

Main amino acids: Proline, glycine and hydroxyproline

Unique triple helix

Found in skin, bones, tendons, cartilage & blood vessels

Natural collagen production slows down with age

Vitamin C is essential for the body to synthesise collagen

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Fat

Macronutrients



- Provides concentrated energy
- Absorbs fat-soluble vitamins (A, D, E, K)
- Protects & insulates organs
- Produces hormones

Recommended daily intake:

Total fats:
25-35% of daily calorie intake

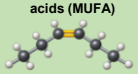
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Fat

Macronutrients

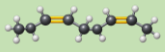
Unsaturated fat

Monounsaturated fatty acids (MUFA)



-C-C=C-C-

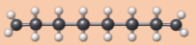
Polyunsaturated fatty acids (PUFA)



-C=C-C-C=C-

- ✓ Contains one or more carbon-carbon double bonds, which put a kink in the chain
- ✓ Prevents tight packing
- ✓ Liquid at room temperature
- ✓ Olive oil, fish oil, avocado

Saturated fat



-C-C-C-C-

- ✓ No double bonds – every carbon is saturated with hydrogen
- ✓ A straight chain that packs tightly together
- ✓ Solid at room temperature
- ✓ Butter, coconut oil, fatty meat

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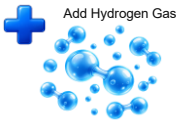
Fat

Macronutrients

Trans-fat



Liquid oil (unsaturated) cis orientation



Add Hydrogen Gas

Partial Hydrogenation



Trans-fatty acids

Source:
Margarine
Fried Food
Processed Food

Why hydrogenation?

1. Converts liquid oil into semi-solid fats
2. Improves texture
3. Improves shelf life

Why bad?

1. Raises LDL (bad cholesterol)
2. Lowers HDL (good cholesterol)

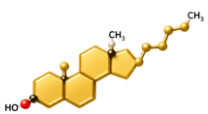
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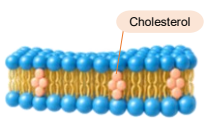
Macronutrients

Cholesterol

- A structural component of **cell membranes**
- A precursor for **bile acids, steroid hormones & vitamin D**



Cholesterol molecule



Cholesterol in cell membrane

Where does cholesterol come from?

1. Body synthesis: **Liver**
2. Diet: Mainly in **animal-derived foods**

Diet sources:



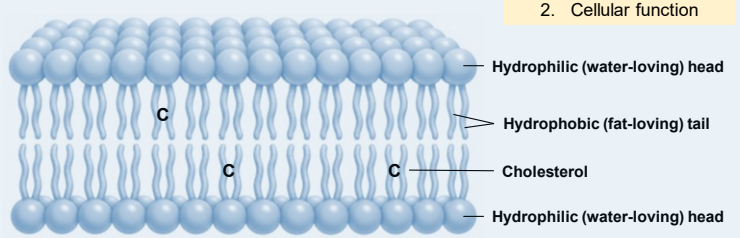
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Fat

Macronutrients

Phospholipid bilayer: Basic structure of cell membrane

Dietary fat affects:
1. Membrane flexibility
2. Cellular function



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What are micronutrients?

- Nutrients needed by the body in SMALL amounts
- Essential for bodily functions: immunity, bone health and energy production

Vitamins

- Organic compounds from plants & animals
- Contains carbon
- Fragile
- Essential for metabolism, immune function and growth



Minerals

- Inorganic elements from soil and water
- No carbon
- Stable
- Essential for bone health, fluids and nerve signals



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Vitamins

Micronutrients

Fat-soluble vitamins
A,D,E,K

- Dissolve in fat
- Absorbed alongside dietary fat
- Stored in the liver and fatty tissue
- Over-supplementation may cause toxicity

Water-soluble vitamins
B,C

- Dissolve in water
- Absorbed directly into the bloodstream
- Not stored (except B12)
- Excess is excreted in urine
- Requires regular intake through diet

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Vitamins

Micronutrients

Vitamin A

Function:
1. Vision
2. Immune function
3. Skin health

Source:
- ✓ Liver
- ✓ Carrot
- ✓ Sweet potato

Provitamin A

- Plant pigments that the body converts into active vitamin A
- Most common: beta-carotene

Source:
- ✓ Carrot
- ✓ Spinach
- ✓ Pumpkin
- ✓ Leafy greens

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Vitamins

Micronutrients

Vitamin D

Function:
1. Bone health
2. Calcium absorption

Source:
- ✓ Sunlight exposure
- ✓ Fish
- ✓ Fortified milk

Vitamin D2 (Ergocalciferol)

- Sourced from plants and fungi
- Fortified food

Vitamin D3 (Cholecalciferol)

- Sourced from animals
- Made in skin from sunlight

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Vitamins

Micronutrients



Vitamin E

Function:

1. Antioxidant
2. Immune function
3. Skin health

Source:

- ✓ Nuts
- ✓ Seeds
- ✓ Vegetable oil

Tocopherols (α, β, γ and δ)

- Saturated side chain
- α-tocopherol: Most biologically active form of Vitamin E

Tocotrienols (α, β, γ and δ)

- Unsaturated side chain
- Exhibit antioxidant function

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Vitamins

Micronutrients



Vitamin K

Function:

1. Blood clotting
2. Bone metabolism

Source:

- ✓ Leafy green vegetables
- ✓ Some fermented food

Vitamin K1 (Phylloquinone)

- ✓ Leafy green vegetables: spinach, kale

Vitamin K2 (Menaquinone)

- Fermented foods: natto, fermented cheese

CAUTION!

Interacts with blood-thinning medication: Warfarin

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Vitamins

Micronutrients



Vitamin B

Function:

1. Nerve function
2. Red blood cell production

Source:

- ✓ Fish
- ✓ Leafy green vegetables
- ✓ Seeds

B1, Thiamine: Nerve health
B2, Riboflavin: Skin, eye and nerve health
B3, Niacin: Nerve and skin health
B5, Pantothenic acid: Fat metabolism
B6, Pyridoxine: Forms haemoglobin, brain health
B7, Biotin: Synthesises fatty acids
B9, Folate: Forms healthy red blood cells, cell growth
B12, Cobalamin: Forms red blood cells, nerve health

B1, B2, B3, B5 and B7 converts food to energy

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Vitamins

Micronutrients



Vitamin C

Function:

1. Antioxidant
2. Collagen synthesis
3. Immunity support

Source:

- ✓ Citrus
- ✓ Guava
- ✓ Vegetables

Ascorbic acid

- Active form, body uses directly

Tips to get more vitamin C from the diet?

1. Eat in fresh: raw fruits and vegetables
2. Avoid boiling, use steaming or microwaving to reduce nutrient loss
3. Avoid prolonged storage and exposure to light, which degrades vitamin C

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Minerals

Micronutrients

Major minerals

Needed in large amount

- Calcium
- Phosphorus
- Magnesium
- Sodium*
- Chloride*
- Potassium*

Trace minerals

Needed in small amounts, but essential

- Iron
- Zinc
- Chromium
- Copper
- Fluoride
- Iodine
- Manganese
- Selenium

* also known as electrolytes

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Minerals

Micronutrients

Ca CALCIUM

~99% of body calcium is stored in bone/teeth, serving as a reservoir

Functions:

1. Bone and teeth strength
2. Muscle contraction
3. Nerve signaling
4. Blood clotting

Source:

- ✓ Dairy
- ✓ Sardines
- ✓ Leafy vegetables



Mg MAGNESIUM

Functions:

1. Energy production
2. Muscle function
3. Nerve function

Source:

- ✓ Nuts
- ✓ Whole grains
- ✓ Dark chocolate



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Minerals

Micronutrients

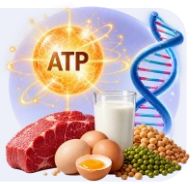
P PHOSPHORUS

Functions:

1. Energy production: Core component of ATP
2. Bone health: Works with calcium
3. Genetic material: Part of DNA and RNA

Source:

- ✓ Meat
- ✓ Dairy
- ✓ Egg
- ✓ Legumes



Zn ZINC

Functions:

1. Immune support
2. Wound healing
3. Cell growth

Source:

- ✓ Red meat
- ✓ Seafoods
- ✓ Nuts
- ✓ Whole grains



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Minerals

Micronutrients

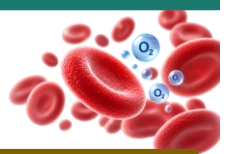
Fe

Iron

Function:

- Transports oxygen in blood
- Prevents anaemia

Is the **core of haemoglobin** (protein in red blood cells) that carries oxygen from the lungs to every tissue in the body



Heme iron

More easily absorbed and used by the body

Animal source:

- ✓ Red meat
- ✓ Poultry
- ✓ Fish



Non-heme iron

Less easily absorbed – to pair with Vitamin C to improve absorption

Plant source:

- ✓ Spinach
- ✓ Lentils
- ✓ Grains



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Minerals

Micronutrients

Electrolytes

Minerals that carry an electric charge when dissolved in the body's fluids

Na Sodium

- Fluid balance
- Nerve signals



Sources:
- Table salt
- Soy sauce
- Processed meat

K Potassium

- Fluid balance
- Heart function
- Muscle contractions



Sources:
- Banana
- Potato
- Yam

Cl Chloride

- Fluid balance
- Digestion: key component of hydrochloric acid in stomach



Sources:
- Table salt
- Seaweed
- Seafood

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Minerals

Micronutrients

Sodium savvy

Limit excess intake: excess intake linked to high blood pressure

Smart swaps:



Compare labels: Sodium content per serving



Season with herbs and spices instead of extra salt



Choose less-processed foods



Cook from scratch rather than relying on packaged meals

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Phytonutrients – Rainbow of health

Phytonutrients (also known as phytochemicals) are natural compounds found in plant-based foods.

Key Benefits:

- Antioxidants protection
- Anti-inflammatory action
- Immune system strengthening
- Lower risk of chronic diseases (heart disease, cancer, diabetes)

Eat a rainbow: consuming a variety of colourful vegetables and fruits increases phytonutrient intake.



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Phytonutrients – Rainbow of health

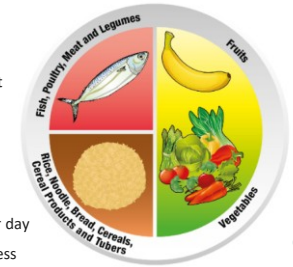


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Hydration — Top up your healthy plate

Why water is ESSENTIAL?

- Regulates body temperature
- Transports nutrients and oxygen to cells
- Removes waste products through urine and sweat
- Lubricates joints and cushions organs
- Supports digestion and kidney function




6-8 glasses

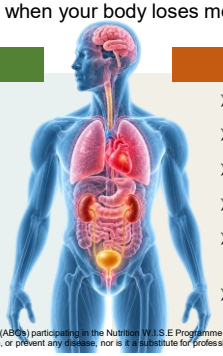
Recommended daily intake:

- Adults: 6-8 glasses (approximately 1.5-2 litres) per day
- More needed during hot weather, exercise or illness
- Signs of dehydration: dark urine, headache, dizziness, fatigue

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Dehydration – Signs and symptoms

Dehydration happens when your body loses more fluid than it takes



Mild / Moderate

- Thirst
- Dry mouth
- Less urinating / Dark yellow urine
- Headache / Dizziness
- Fatigue

Severe

- Extreme thirst
- Rapid heartbeat
- Rapid breathing
- Sunken eyes
- Not urinating / very dark yellow / amber-coloured urine
- Irritability or confusion / unconsciousness / delirium

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Hydration

Diuretics

Increase urine output, causing the body to lose more fluid than usual

Dose-dependent diuretics

- ✓ Caffeine (coffee, tea, energy drink)
- ✓ Herbal teas (Hibiscus tea, dandelion, parsley)
- ✓ Alcohol

Coffee and tea have a mild diuretic effect, but moderate intake still contributes positively to daily hydration overall

Insensible water loss

- Water loss without noticing/sweating
- Increases with fever, high altitude, hot/dry environment

Breathing

Water loss through exhalation

Skin

Water evaporates from the skin even without visible sweat



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